

Ensuring Singapore's role as a refining and exporting hub in ASEAN

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Date: 18 September 2026

Subject: Ensure Singapore's continued function as a refining hub in ASEAN

Executive Summary: Singapore's dependence on Middle Eastern crude oil exposes its refining and refined-fuel export hub to geopolitical disruptions, threatening domestic energy security and ASEAN's fuel supply resilience. The Strait of Hormuz crisis highlights vulnerabilities arising from concentrated crude sourcing, limited refinery feedstock flexibility, and downstream regional dependence. This memo evaluates three policy options: strengthening strategic petroleum buffers, diversifying crude processing capabilities, and establishing a refined-fuel allocation framework. It recommends that Singapore use its ASEAN Chairmanship to operationalise a regional petroleum stockpile and establish pre-agreed export allocation rules, strengthening collective resilience while preserving Singapore's role as a reliable regional refining and export hub.

Problem Statement: With the closure of the Strait of Hormuz since February 2026, [crude oil prices have now crossed US \\$100 per barrel](#), causing a double effect of both rising prices and tighter physical supplies. For Singapore, this is linked directly to the country's crude oil import reliance on the Middle East. Amidst the reduced inflow of crude from the Gulf, concerns emerge about ensuring Singapore's role as a trading and exporting hub of refined fuels for different markets, especially for the neighbouring ASEAN countries

Background:

On 28 February 2026, Iran closed the Strait of Hormuz to commercial traffic, owing to the US and Israeli Military Strikes. As a result, commercial vessel transits through the Strait have fallen to [single digits per day](#), down from pre-conflict levels of up to 125 vessels daily. Prior to the Iran war, this route accounted for about 20 per cent of the world's daily crude oil and Liquefied Natural Gas Supply.

This served a crucial role for the rest of the world, including Singapore that has established itself as the [fifth largest refinery and export hub](#) in the world. This drives its significantly high oil consumption of 1.4 million barrels per day. Its three key refineries, Aster, ExxonMobil and Singapore Refining Co., together possess a refining capacity of 1.27 million barrels per day.

Owing to its continued role as a refining hub, Singapore imports 100% of the crude oil it uses, largely from the Middle East. Recent trade data has shown that [73.5% of Singapore's crude oil imports](#) come from five Middle Eastern countries: the UAE, Qatar, Iraq, Saudi Arabia and Oman, while the UAE alone accounts for 43% of this.

Analysis:

Concentrated upstream exposure to the Gulf:

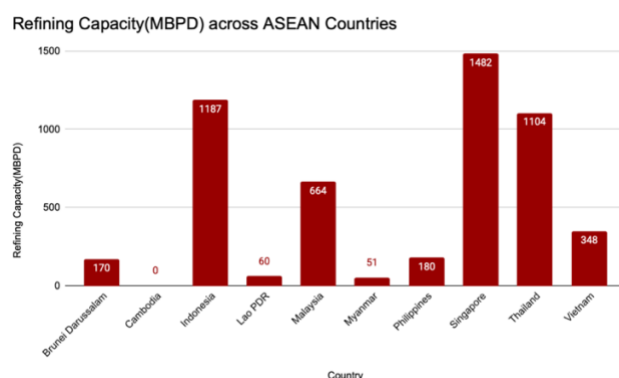
The refineries based in Singapore are tailored to process a [medium-sour oil mix](#), with Middle Eastern crude oil accounting for 70 per cent of it. With the 2025 refinery expansion of ExxonMobil in Singapore, this dependency on Gulf crude increased even further. Estimates show that although refineries can switch 15 to 30 per cent of their feed to alternative fuels, [they will not run optimally](#). As a

result, two major Singapore petrochemical firms, PCS (Petrochemical Corporation of Singapore) and Aster Chemicals and Energy, have [announced force majeure declarations](#) due to feedstock disruptions stemming from the Strait of Hormuz blockade.

Moreover, owing to refinery run cuts during this time, refineries have been operating at around 50-60% capacity. This slowdown has been a key driver of a contraction in manufacturing activity. Petroleum and petrochemicals, that form the chemical cluster, account for about 15% of total manufacturing output and around [1.7% of GDP](#). Hence, disruptions in refinery running capacity have affected other aspects of the Singaporean economy, including its [growth outlook](#).

Exposure Downstream

The upstream challenge of Singapore's reliance on Middle Eastern crude is further aggravated by the fact that it acts as a feedstock for neighbouring countries that import refined fuel from it. Refining capacity in ASEAN has remained concentrated amongst a few countries, with Singapore acting as the possessing the highest capacity in the region, of [1482 Million Barrels per day](#).



Source: ASEAN Centre for Energy

This concentration matters because ASEAN's exposure to the Hormuz shocks is not uniform. While Singapore, as an import refiner, can partly cushion the shock through reserve depths, countries that import already refined products from crude oil have no such buffer and [suffer the brunt](#) of availability shocks. Countries such as Myanmar, Indonesia, Vietnam and Cambodia depend on Singapore to meet a [major portion](#) of their refined fuel needs. Especially during times of crisis such as the current geopolitical tensions, the vulnerability of countries such as Laos and Cambodia, which do not possess refining capacity and rely on external imports, is further exposed. In May this year, Cambodia had to [export more fuel from Singapore](#) and Malaysia due to fuel export restrictions by China and Vietnam. Singapore's continued function as a reliable refining and export hub is therefore not a matter of its own energy security alone but also a regional responsibility underpinning ASEAN's collective resilience to supply shocks.

Policy Options: These are a few categorical options that Singapore can use in better preparing itself for future Geopolitical shocks.

Policy Option 1: Reduce Supply Dependence & build strategic buffers:

- **Pro:** Directly reduces supply concentration risk by providing a buffer against oil price volatility and disruptions, without requiring immediate changes in supplier behaviour.
- **Con:** Alternative crude sources may be hard to secure in the short term, may be of lower quality, and might require new shipping and insurance arrangements. Stockpiles also entail high costs, including storage, tied-up capital, and product degradation.

Policy Option 2: Diversify Crude Processing Capabilities

Make long-term investment in refineries that can process different crude qualities.

- **Pro:** Addresses the underlying technical constraint of only being able to mix 15-30% of other oil, by enhancing long-term structural resilience.
- **Con:** Capital-intensive and time-consuming, with retrofits potentially taking several years to complete, limiting their effectiveness against near-term supply shocks. Implementation would also require co-funding negotiations with separate private operators whose investment incentives may not be aligned.

Policy Option 3: Building an oil disbursement prioritization framework

- **Pro:** Establishes clear, pre-agreed rules for allocation of limited supplies, reducing ad hoc decision-making during times of crisis conditions.
- **Con:** The policy option is sensitive to negotiation, especially since Singapore also caters to countries such as Australia with respect to exporting refined fuel products.

Recommendations:

1. **Supply Side Measure: Push for the regional Oil Stockpile as it becomes the ASEAN Chair:**
Singapore should use its incoming ASEAN Chairmanship to push for the regional joint oil stockpile provision, which already exists in the ASEAN petroleum security, but has not been operationalised. Most ASEAN countries hold an [average of 20 to 50 days of oil stock](#). Moreover, with the Strait of Hormuz disruption, importing oil and gas at current crisis prices increased the region's import costs by [US\\$3.36 billion per month above 2026 expectations](#). The existence of a rice stockpile reserve under the ASEAN plus emergency rice reserve provides an existing example in place.
2. **Demand Side Measure: Pre-agreed regional export allocation framework:**
Rather than decide on a case-by-case basis, Singapore should fix allocation rules and shortage thresholds in advance. Such a pre-agreed protocol would further align with [APSA's own crisis protocol](#), which requires members to demonstrate that they have implemented demand restraint measures before regional assistance.